



EXTENSIONS OF THE TWO-VARIABLE LINEAR REGRESSION MODEL

6.1 REGRESSION THROUGH THE ORIGIN

- Lets take a two-variable population regression equation assuming the following form:

$$Y_i = \beta_2 X_i + u_i$$

- In this model the intercept term is absent or zero and this is named as **regression through the origin**.
- As an illustration, consider the capital asset pricing model (CAPM) of modern portfolio theory, which, in its risk-premium form, may be expressed as

$$(ER_i - r_f) = \beta_i(ER_m - r_f)$$

- where ER_i = expected rate of return on security i
 ER_m = expected rate of return on the market portfolio as represented by, say, the S&P 500 composite stock index

REGRESSION THROUGH THE ORIGIN

- r_f = risk-free rate of return, like the return on 90-day Treasury bills
- β_i = the Beta coefficient, a measure of systematic risk, i.e., risk that cannot be eliminated through diversification. Also, a measure of the extent to which the i th security's rate of return moves with the market.
- A $\beta_i > 1$ implies a volatile or aggressive security, whereas a $\beta_i < 1$ suggests a defensive security.
- (*Note:* Do not confuse this β_i with the slope coefficient of the twovariable regression, β_2 .)
- If capital markets work efficiently, then CAPM postulates that security i 's expected risk premium ($= ER_i - r_f$) is equal to that security's β coefficient times the expected market risk premium ($= ER_m - r_f$).

REGRESSION THROUGH THE ORIGIN

- If the CAPM holds, we have the situation depicted in Figure 6.1. The line shown in the figure is known as the **security market line** (SML).
- For empirical purposes, Equation 6.1.2 is often expressed as

$$R_i - r_f = \beta_i(R_m - r_f) + u_i$$

- Or

$$R_i - r_f = \alpha_i + \beta_i(R_m - r_f) + u_i$$

- The latter model is known as the **Market Model**. If CAPM holds, α_i is expected to be zero. (See Figure 6.2.)

REGRESSION THROUGH THE ORIGIN

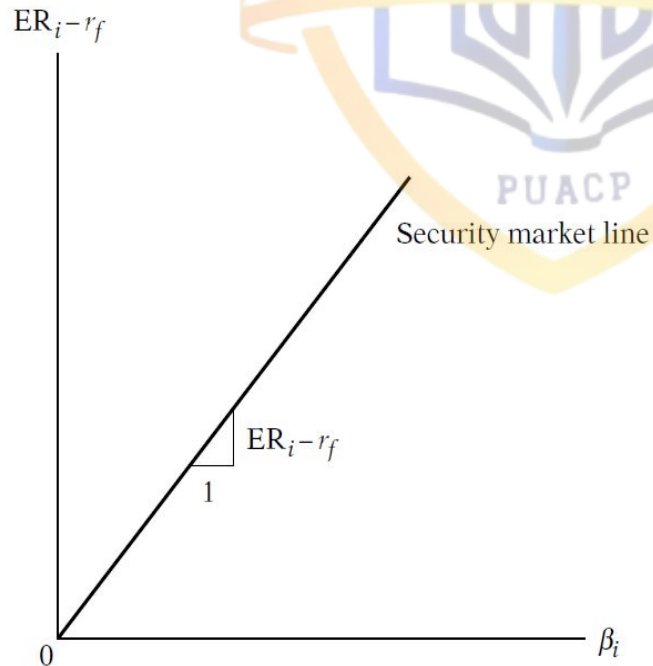


FIGURE 6.1 Systematic risk.

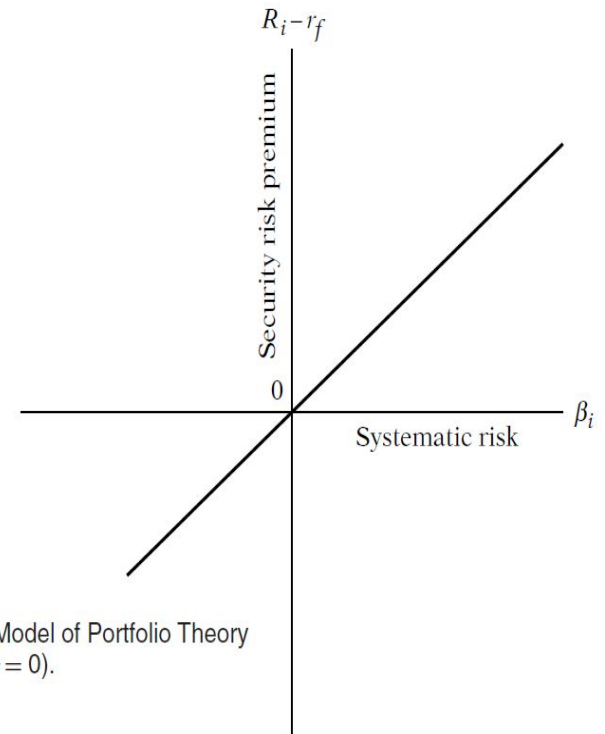


FIGURE 6.2 The Market Model of Portfolio Theory (assuming $\alpha_i = 0$).

REGRESSION THROUGH THE ORIGIN

How do we estimate models like Eq. (6.1.1), and what special problems do they pose? To answer these questions, let us first write the sample regression function (SRF) of Eq. (6.1.1), namely,

$$Y_i = \hat{\beta}_2 X_i + \hat{u}_i$$

Now applying the ordinary least squares (OLS) method to Eq. (6.1.5), we obtain the following formulas for $\hat{\beta}_2$ and its variance (proofs are given in Appendix 6A, Section 6A.1):

$$\hat{\beta}_2 = \frac{\sum X_i Y_i}{\sum X_i^2}$$
$$\text{var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum X_i^2}$$

where σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n - 1}$$

REGRESSION THROUGH THE ORIGIN

It is interesting to compare these formulas with those obtained when the intercept term is included in the model:

$$\hat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2}$$

$$\text{var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2}$$

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n - 2}$$

the conventional r^2 given in Chapter 3 is not appropriate for regressions that do not contain the intercept. But one can compute what is known as the **raw r^2** for such models, which is defined as

$$\text{raw } r^2 = \frac{(\sum X_i Y_i)^2}{\sum X_i^2 \sum Y_i^2}$$

Note: These are raw (i.e., not mean-corrected) sums of squares and cross products. Although this raw r^2 satisfies the relation $0 < r^2 < 1$, it is not directly comparable to the conventional r^2 value. For this reason some authors do not report the r^2 value for zero intercept regression models.

REGRESSION THROUGH THE ORIGIN

- Monthly data for the period 1980-1999 for total 240 observations is provided in Table 6.1.
- Y_t refers to excess returns on index of 1034 stocks in the consumer good sector, while X_t indicates excess returns on the overall stock market index.

Dependent Variable: Y
Method: Least Squares
Sample: 1980M01 1999M12
Included observations: 240

	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.447481	0.362943	-1.232924	0.2188
X	1.171128	0.075386	15.53500	0.0000
R-squared	0.503480	Mean dependent var.		0.499826
Adjusted R-squared	0.501394	S.D. dependent var.		7.849594
S.E. of regression	5.542759	Durbin-Watson stat.		1.984746
Sum squared resid.	7311.877	Prob. (F-statistic)		0.000000
F-statistic	241.3363			

REGRESSION THROUGH THE ORIGIN

- As these results show, the slope coefficient, which is the Beta coefficient, is highly significant, for its p value is extremely small.
- It shows that if the excess market rate goes up by 1 percentage point, the excess return on the index of consumer goods sector goes up by about 1.15 percentage points. Not only is the slope coefficient statistically significant, but it is significantly greater than 1 (can you verify this?). If a Beta coefficient is greater than 1, such a security (here a portfolio of 104 stocks) is said to be **volatile**; it moves more than proportionately with the overall stock market index.
- By the way, note that the r^2 value given for the regression-through-the-origin model should be taken with a grain of salt, for the traditional formula of r^2 is not applicable for such models. *EViews*, however, routinely presents the standard r^2 value even for such models.

SCALING AND UNITS OF MEASUREMENT

TABLE 6.2

Gross Private Domestic Investment and GDP, United States, 1990–2005
(Billions of chained [2000] dollars, except as noted; quarterly data at seasonally adjusted annual rates)

Source: *Economic Report of the President*, 2007, Table B-2, p. 328.

Year	GPDIBL	GPDIM	GDPB	GDPM
1990	886.6	886,600.0	7,112.5	7,112,500.0
1991	829.1	829,100.0	7,100.5	7,100,500.0
1992	878.3	878,300.0	7,336.6	7,336,600.0
1993	953.5	953,500.0	7,532.7	7,532,700.0
1994	1,042.3	1,042,300.0	7,835.5	7,835,500.0
1995	1,109.6	1,109,600.0	8,031.7	8,031,700.0
1996	1,209.2	1,209,200.0	8,328.9	8,328,900.0
1997	1,320.6	1,320,600.0	8,703.5	8,703,500.0
1998	1,455.0	1,455,000.0	9,066.9	9,066,900.0
1999	1,576.3	1,576,300.0	9,470.3	9,470,300.0
2000	1,679.0	1,679,000.0	9,817.0	9,817,000.0
2001	1,629.4	1,629,400.0	9,890.7	9,890,700.0
2002	1,544.6	1,544,600.0	10,048.8	10,048,800.0
2003	1,596.9	1,596,900.0	10,301.0	10,301,000.0
2004	1,713.9	1,713,900.0	10,703.5	10,703,500.0
2005	1,842.0	1,842,000.0	11,048.6	11,048,600.0

Note: GPDIBL = gross private domestic investment, billions of 2000 dollars.
 GPDIM = gross private domestic investments, millions of 2000 dollars.
 GDPB = gross domestic product, billions of 2000 dollars.
 GDPM = gross domestic product, millions of 2000 dollars.

- Consider the data given in Table 6.2, which refers to U.S. gross private domestic investment (GPDI) and gross domestic product (GDP), in billions as well as millions of (chained) 2000 dollars.

SCALING AND UNITS OF MEASUREMENT

Suppose in the regression of GPD_I on GDP one researcher uses data in billions of dollars but another expresses data in millions of dollars. Will the regression results be the same in both cases? If not, which results should one use? In short, do the units in which the regressand and regressor(s) are measured make any difference in the regression results? If so, what is the sensible course to follow in choosing units of measurement for regression analysis? To answer these questions, let us proceed systematically. Let

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i \quad (6.2.1)$$

where $Y = \text{GPD}_I$ and $X = \text{GDP}$. Define

$$Y_i^* = w_1 Y_i \quad (6.2.2)$$

$$X_i^* = w_2 X_i \quad (6.2.3)$$

where w_1 and w_2 are constants, called the **scale factors**; w_1 may equal w_2 or be different.

SCALING AND UNITS OF MEASUREMENT

EXAMPLE 6.2

*The Relationship
between the
GDPI and GDP,
United States,
1990–2005*

To substantiate the preceding theoretical results, let us return to the data given in Table 6.2 and examine the following results (numbers in parentheses are the estimated standard errors).

Both GDPI and GDP in billions of dollars:

$$\begin{aligned}\widehat{\text{GDPI}}_t &= -926.090 + 0.2535 \text{ GDP}_t \\ \text{se} &= (116.358) \quad (0.0129) \quad r^2 = 0.9648\end{aligned}\tag{6.2.21}$$

Both GDPI and GDP in millions of dollars:

$$\begin{aligned}\widehat{\text{GDPI}}_t &= -926,090 + 0.2535 \text{ GDP}_t \\ \text{se} &= (116,358) \quad (0.0129) \quad r^2 = 0.9648\end{aligned}\tag{6.2.22}$$

Notice that the intercept as well as its standard error is 1000 times the corresponding values in the regression (6.2.21) (note that $w_1 = 1000$ in going from billions to millions of dollars), but the slope coefficient as well as its standard error is unchanged, in accordance with the theory.

GDPI in billions of **dollars** and GDP in millions of dollars:

$$\begin{aligned}\widehat{\text{GDPI}}_t &= -926.090 + 0.0002535 \text{ GDP}_t \\ \text{se} &= (116.358) \quad (0.0000129) \quad r^2 = 0.9648\end{aligned}\tag{6.2.23}$$

As expected, the slope coefficient as well as its standard error is 1/1000 its value in Eq. (6.2.21), since only the X, or GDP, scale is changed.

GDPI in millions of dollars and GDP in billions of dollars:

$$\begin{aligned}\widehat{\text{GDPI}}_t &= -926,090 + 253.524 \text{ GDP}_t \\ \text{se} &= (116,358.7) \quad (12.9465) \quad r^2 = 0.9648\end{aligned}\tag{6.2.24}$$

SCALING AND UNITS OF MEASUREMENT

From (6.2.2) and (6.2.3) it is clear that Y_i^* and X_i^* are *rescaled* Y_i and X_i . Thus, if Y_i and X_i are measured in billions of dollars and one wants to express them in millions of dollars, we will have $Y_i^* = 1000 Y_i$ and $X_i^* = 1000 X_i$; here $w_1 = w_2 = 1000$.

Now consider the regression using Y_i^* and X_i^* variables:

$$Y_i^* = \hat{\beta}_1^* + \hat{\beta}_2^* X_i^* + \hat{u}_i^*$$

where $Y_i^* = w_1 Y_i$, $X_i^* = w_2 X_i$, and $\hat{u}_i^* = w_1 \hat{u}_i$. (Why?)

We want to find out the relationships between the following pairs:

1. $\hat{\beta}_1$ and $\hat{\beta}_1^*$
2. $\hat{\beta}_2$ and $\hat{\beta}_2^*$
3. $\text{var}(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_1^*)$
4. $\text{var}(\hat{\beta}_2)$ and $\text{var}(\hat{\beta}_2^*)$
5. $\hat{\sigma}^2$ and $\hat{\sigma}^{*2}$
6. r_{xy}^2 and $r_{x^*y^*}^2$

SCALING AND UNITS OF MEASUREMENT

$$\hat{\beta}_1^* = \bar{Y}^* - \hat{\beta}_2^* \bar{X}^* \qquad \hat{\beta}_1^* = \bar{Y}^* - \hat{\beta}_2^* \bar{X}^*$$

$$\hat{\beta}_2^* = \frac{\sum x_i^* y_i^*}{\sum x_i^{*2}}$$

$$\hat{\beta}_2^* = \frac{\sum x_i^* y_i^*}{\sum x_i^{*2}}$$

$$\text{var}(\hat{\beta}_1^*) = \frac{\sum X_i^{*2}}{n \sum x_i^{*2}} \cdot \sigma^{*2}$$

$$\text{var}(\hat{\beta}_1^*) = \frac{\sum X_i^{*2}}{n \sum x_i^{*2}} \cdot \sigma^{*2}$$

$$\text{var}(\hat{\beta}_2^*) = \frac{\sigma^{*2}}{\sum x_i^{*2}}$$

$$\text{var}(\hat{\beta}_2^*) = \frac{\sigma^{*2}}{\sum x_i^{*2}}$$

$$\hat{\sigma}^{*2} = \frac{\sum \hat{u}_i^{*2}}{(n-2)}$$

$$\hat{\sigma}^{*2} = \frac{\sum \hat{u}_i^{*2}}{(n-2)}$$

SCALING AND UNITS OF MEASUREMENT

From these results it is easy to establish relationships between the two sets of parameter estimates. All that one has to do is recall these definitional relationships: $Y_i^* = w_1 Y_i$ (or $y_i^* = w_1 y_i$); $X_i^* = w_2 X_i$ (or $x_i^* = w_2 x_i$); $\hat{u}_i^* = w_1 \hat{u}_i$; $\bar{Y}^* = w_1 \bar{Y}$ and $\bar{X}^* = w_2 \bar{X}$. Making use of these definitions, the reader can easily verify that

$$\hat{\beta}_2^* = \left(\frac{w_1}{w_2} \right) \hat{\beta}_2$$

$$\hat{\beta}_1^* = w_1 \hat{\beta}_1$$

$$\hat{\sigma}^{*2} = w_1^2 \hat{\sigma}^2$$

$$\text{var}(\hat{\beta}_1^*) = w_1^2 \text{var}(\hat{\beta}_1)$$

$$\text{var}(\hat{\beta}_2^*) = \left(\frac{w_1}{w_2} \right)^2 \text{var}(\hat{\beta}_2)$$

$$r_{xy}^2 = r_{x^*y^*}^2$$

6.3 REGRESSION ON STANDARDIZED VARIABLES

- The units in which the regressand and regressor(s) are expressed affect the interpretation of the regression coefficients. This can be avoided if we are willing to express the regressand and regressor(s) as standardized variables (Mean zero, variance 1).
- A variable is said to be standardized if we subtract the mean value of the variable from its individual values and divide the difference by the standard deviation of that variable.

Thus, in the regression of Y and X , if we redefine these variables as

$$Y_i^* = \frac{Y_i - \bar{Y}}{S_Y} \quad (6.3.1)$$

$$X_i^* = \frac{X_i - \bar{X}}{S_X} \quad (6.3.2)$$

where $\bar{Y}$ = sample mean of Y , S_Y = sample standard deviation of Y , $\bar{X}$ = sample mean of X , and S_X is the sample standard deviation of X ; the variables Y_i^* and X_i^* are called **standardized variables**.

To show the difference between Eqs. (6.3.3) and (6.3.5), let us return to the GPDI and GDP example discussed in the preceding section. The results of (6.2.21) discussed previously are reproduced here for convenience.

$$\widehat{\text{GPDI}}_t = -926.090 + 0.2535 \text{ GDP}_t \quad (6.3.6)$$

$$\text{se} = (116.358) \quad (0.0129) \quad r^2 = 0.9648$$

where GPDI and GDP are measured in billions of dollars.

The results corresponding to Eq. (6.3.5) are as follows, where the starred variables are standardized variables:

$$\widehat{\text{GPDI}}_t^* = 0.9822 \text{ GDP}_t^* \quad (6.3.7)$$

$$\text{se} = (0.0485)$$

We know how to interpret Eq. (6.3.6): If GDP goes up by a dollar, on average GPDI goes up by about 25 cents. How about Eq. (6.3.7)? Here the interpretation is that if the (standardized) GDP increases by one standard deviation, on average, the (standardized) GPDI increases by about 0.98 standard deviations.

Note: Refer to page 158, please read out the advantages of standardized regression in book please.

FUNCTIONAL FORMS OF REGRESSION MODELS

- This text is concerned primarily with models that are linear in the parameters; they may or may not be linear in the variables.
- In the sections that follow we consider some commonly used regression models that may be nonlinear in the variables but are linear in the parameters or that can be made so by suitable transformations of the variables.
- In particular, we discuss the following regression models:
 - 1. The log-linear model
 - 2. Semilog models
 - 3. Reciprocal models
 - 4. The logarithmic reciprocal model

We discuss the special features of each model, when they are appropriate, and how they are estimated. Each model is illustrated with suitable examples.

HOW TO MEASURE ELASTICITY: THE LOG-LINEAR MODEL

Consider the following model, known as the **exponential regression model**:

$$Y_i = \beta_1 X_i^{\beta_2} e^{u_i}$$

which may be expressed alternatively as

$$\ln Y_i = \ln \beta_1 + \beta_2 \ln X_i + u_i$$

where $\ln$ = natural log (i.e., log to the base e , and where $e = 2.718$).

If we write above equation) as

$$\ln Y_i = \alpha + \beta_2 \ln X_i + u_i$$

where $\alpha = \ln \beta_1$, this model is linear in the parameters α and β_2 , linear in the logarithms of the variables Y and X , and can be estimated by OLS regression. Because of this linearity, such models are called **log-log**, **double-log**, or **log-linear** models.

HOW TO MEASURE ELASTICITY: THE LOG-LINEAR MODEL

If the assumptions of the classical linear regression model are fulfilled, the parameters of this equation can be estimated by the OLS method by letting

$$Y_i^* = \alpha + \beta_2 X_i^* + u_i$$

where $Y_i^* = \ln Y_i$ and $X_i^* = \ln X_i$. The OLS estimators $\hat{\alpha}$ and $\hat{\beta}_2$ obtained will be best linear unbiased estimators of α and β_2 , respectively.

One attractive feature of the log-log model, which has made it popular in applied work, is that the slope coefficient β_2 measures the elasticity of Y with respect to X , that is, the per-centage change in Y for a given (small) percentage change in X . Thus, if Y represents the quantity of a commodity demanded and X its unit price, β_2 measures the price elasticity of demand

Example for The Log-Linear Model

EXAMPLE 6.3

*Expenditure
on Durable
Goods in
Relation to
Total Personal
Consumption
Expenditure*

Table 6.3 presents data on total personal consumption expenditure (PCEXP), expenditure on durable goods (EXPDUR), expenditure on nondurable goods (EXPNONDUR), and expenditure on services (EXPSERVICES), all measured in 2000 billions of dollars.¹³

Suppose we wish to find the elasticity of expenditure on durable goods with respect to total personal consumption expenditure. Plotting the log of expenditure on durable goods against the log of total personal consumption expenditure, you will see that the relationship between the two variables is linear. Hence, the double-log model may be appropriate. The regression results are as follows:

$$\begin{aligned}\widehat{\ln \text{EXPDUR}_t} &= -7.5417 + 1.6266 \ln \text{PCEXP}_t \\ \text{se} &= (0.7161) \quad (0.0800) \\ t &= (-10.5309)^* \quad (20.3152)^* \quad r^2 = 0.9695\end{aligned}\tag{6.5.5}$$

where * indicates that the p value is extremely small.

TABLE 6.3

**Total Personal
Expenditure and
Categories
(Billions of chained
[2000] dollars;
quarterly data at
seasonally adjusted
annual rates)**

Year or quarter	EXPSERVICES	EXPDUR	EXPNONDUR	PCEXP
2003-I	4,143.3	971.4	2,072.5	7,184.9
2003-II	4,161.3	1,009.8	2,084.2	7,249.3
2003-III	4,190.7	1,049.6	2,123.0	7,352.9
2003-IV	4,220.2	1,051.4	2,132.5	7,394.3
2004-I	4,268.2	1,067.0	2,155.3	7,479.8
2004-II	4,308.4	1,071.4	2,164.3	7,534.4
2004-III	4,341.5	1,093.9	2,184.0	7,607.1
2004-IV	4,377.4	1,110.3	2,213.1	7,687.1
2005-I	4,395.3	1,116.8	2,241.5	7,739.4
2005-II	4,420.0	1,150.8	2,268.4	7,819.8
2005-III	4,454.5	1,175.9	2,287.6	7,895.3
2005-IV	4,476.7	1,137.9	2,309.6	7,910.2
2006-I	4,494.5	1,190.5	2,342.8	8,003.8

Sources: Department of
Commerce, Bureau of
Economic Analysis.
*Economic Report
of the President, 2007,*
Table B-17, p. 347.

SEMILOG MODELS: LOG-LIN & LIN-LOG MODEL

- These models are relevant to find out rate of growth of certain variables in economics, such as population, money supply, GNP, employment etc.
- Let Y_t denote real expenditure on services at time t and Y_0 the initial value of the expenditure on services. According to well-known compound interest formula from your introductory course in economics.

$$Y_t = Y_0(1 + r)^t$$

where r is the compound (i.e., over time) rate of growth of Y . Taking the natural logarithm of Equation, we can write

$$\ln Y_t = \ln Y_0 + t \ln(1 + r)$$

Now letting

$$\beta_1 = \ln Y_0$$

$$\beta_2 = \ln(1 + r)$$

we can write Equation as

$$\ln Y_t = \beta_1 + \beta_2 t$$

Adding the disturbance term to the above equation, we obtain

$$\ln Y_t = \beta_1 + \beta_2 t + u_t$$

This model is like any other linear regression model in that the parameters β_1 and β_2 are linear. The only difference is that the regressand is the logarithm of Y and the regressor is “time,” which will take values of 1, 2, 3, etc.

SEMILOG MODELS: LOG-LIN & LIN-LOG MODEL

In this model *the slope coefficient measures the constant proportional or relative change in Y for a given absolute change in the value of the regressor* (in this case the variable t), that is,

$$\beta_2 = \frac{\text{relative change in regressand}}{\text{absolute change in regressor}}$$

If we multiply the relative change in Y by 100, above equation will then give the percentage change, or the *growth rate*, in Y for an absolute change in X , the regressor. That is, 100 times β_2 gives the growth rate in Y ; 100 times β_2 is known in the literature as the **semielasticity** of Y with respect to X .

Example for Log–Lin Model

EXAMPLE 6.4

*The Rate of
Growth
Expenditure on
Services*

To illustrate the growth model (6.6.6), consider the data on expenditure on services given in Table 6.3. The regression results over time (t) are as follows:

$$\begin{aligned}\widehat{\ln EXS_t} &= 8.3226 + 0.00705t \\ \text{se} &= (0.0016) \quad (0.00018) \quad r^2 = 0.9919 \\ t &= (5201.625)^* \quad (39.1667)^*\end{aligned}\tag{6.6.8}$$

Note: EXS stands for expenditure on services and * denotes that the p value is extremely small.

This equation indicates that over the quarterly period 2003-I to 2006-III, expenditures on services increased at the (quarterly) rate of 0.705 percent.

LIN-LOG MODEL

Unlike the growth model just discussed, in which we were interested in finding the percent growth in Y for an absolute change in X , suppose we now want to find the absolute change in Y for a percent change in X . A model that can accomplish this purpose can be written as:

$$Y_i = \beta_1 + \beta_2 \ln X_i + u_i$$

For descriptive purposes we call such a model a **lin-log model**.

Let us interpret the slope coefficient β_2 . As usual,

$$\begin{aligned}\beta_2 &= \frac{\text{change in } Y}{\text{change in } \ln X} \\ &= \frac{\text{change in } Y}{\text{relative change in } X}\end{aligned}$$

The second step follows from the fact that *a change in the log of a number is a relative change*.

LIN-LOG MODEL

Symbolically, we have

$$\beta_2 = \frac{\Delta Y}{\Delta X/X} \quad (6.6.12)$$

where, as usual, Δ denotes a small change. Equation (6.6.12) can be written, equivalently, as

$$\Delta Y = \beta_2(\Delta X/X) \quad (6.6.13)$$

This equation states that the absolute change in Y ($= \Delta Y$) is equal to slope times the relative change in X . If the latter is multiplied by 100, then (6.6.13) gives the absolute change in Y for a percentage change in X . Thus, if $(\Delta X/X)$ changes by 0.01 unit (or 1 percent), the absolute change in Y is $0.01(\beta_2)$; if in an application one finds that $\beta_2 = 500$, the absolute change in Y is $(0.01)(500) = 5.0$. Therefore, when regression (6.6.11) is estimated by OLS, do not forget to multiply the value of the estimated slope coefficient by 0.01, or, what amounts to the same thing, divide it by 100. *If you do not keep this in mind, your interpretation in an application will be highly misleading.*

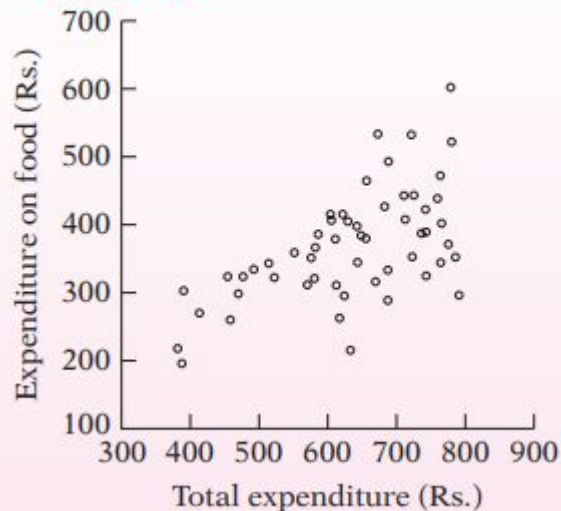
Example for Lin-Log Models

EXAMPLE 6.5

As an illustration of the lin-log model, let us revisit our example on food expenditure in India, Example 3.2. There we fitted a linear-in-variables model as a first approximation. But if we plot the data we obtain the plot in Figure 6.5. As this figure suggests, food expenditure increases more slowly as total expenditure increases, perhaps giving credence to Engel's law. The results of fitting the lin-log model to the data are as follows:

$$\widehat{\text{FoodExp}}_i = -1283.912 + 257.2700 \ln \text{TotalExp}_i$$
$$t = \quad (-4.3848)^* \quad (5.6625)^* \quad r^2 = 0.3769 \quad (6.6.14)$$

FIGURE 6.5



Interpreted in the manner described earlier, the slope coefficient of about 257 means that an increase in the total food expenditure of 1 percent, on average, leads to about 2.57 rupees increase in the expenditure on food of the 55 families included in the sample. (Note: the estimated slope coefficient is divided by 100.)

RECIPROCAL MODELS

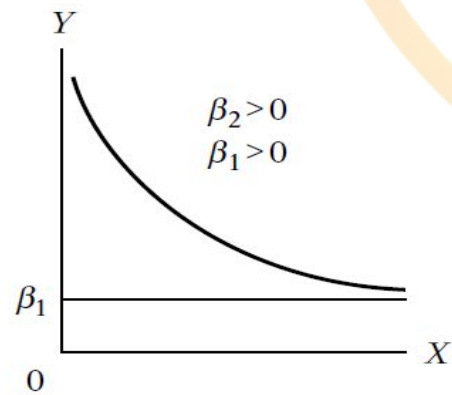
Models of the following type are known as **reciprocal** models.

$$Y_i = \beta_1 + \beta_2 \left(\frac{1}{X_i} \right) + u_i \quad (6.7.1)$$

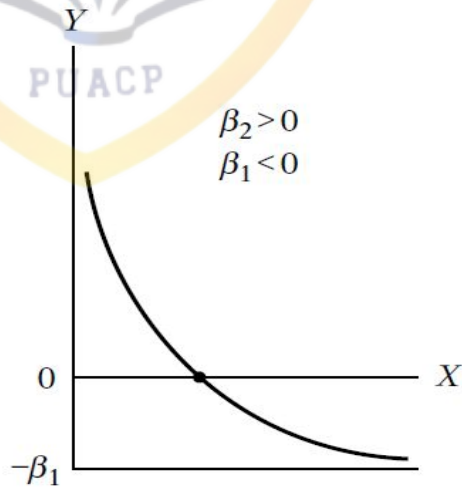
Although this model is nonlinear in the variable X because it enters inversely or reciprocally, the model is linear in β_1 and β_2 and is therefore a linear regression model.

This model has these features: As X increases indefinitely, the term $\beta_2(1/X)$ approaches zero (*note: β_2 is a constant*) and Y approaches the limiting or *asymptotic* value β_1 . Therefore, models like (6.7.1) have built in them an **asymptote** or limit value that the dependent variable will take when the value of the X variable increases indefinitely.

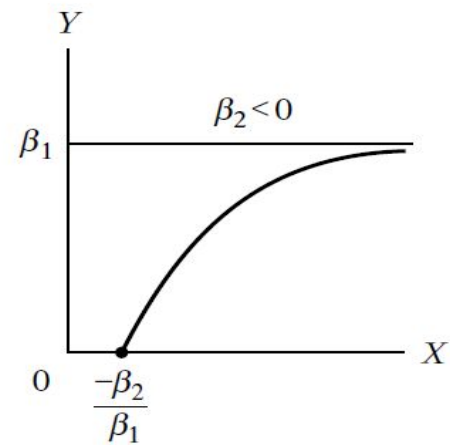
RECIPROCAL MODELS



(a)



(b)

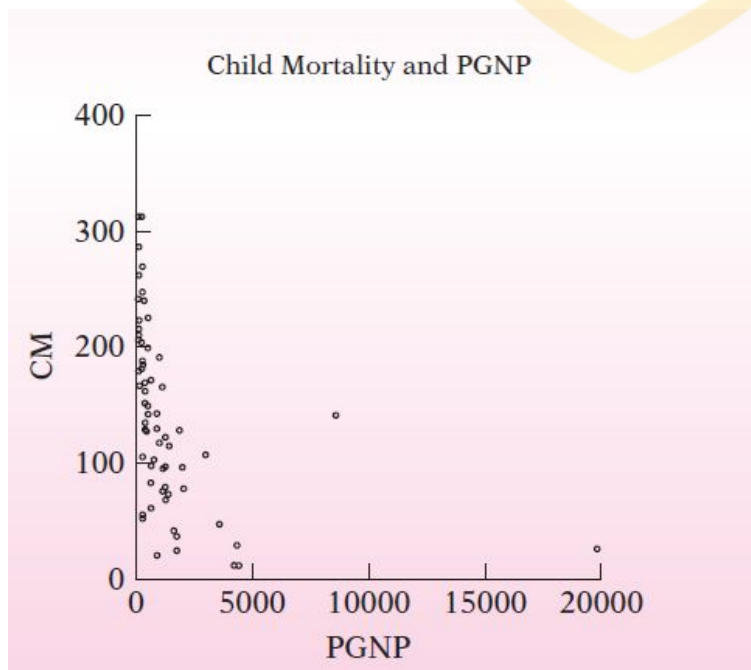


(c)

The reciprocal model: $Y = \beta_1 + \beta_2 \left(\frac{1}{X} \right)$.

RECIPROCAL MODELS

Example 6.6 : Relationship between child mortality and per capita GNP in 66 countries



As per capita GNP increases, one would expect child mortality to decrease because people can afford to spend more on health care, assuming all other factors remain constant. But the relationship is not a straight line one: As per capita GNP increases, initially there is a dramatic drop in CM but the drop tapers off as per capita GNP continues to increase.

RECIPROCAL MODELS

If we try to fit the reciprocal model (6.7.1), we obtain the following regression results:

$$\begin{aligned}\widehat{CM}_i &= 81.79436 + 27,237.17 \left(\frac{1}{PGNP_i} \right) \\ \text{se} &= (10.8321) \quad (3759.999) \\ t &= (7.5511) \quad (7.2535) \quad r^2 = 0.4590\end{aligned}\tag{6.7.2}$$

As per capita GNP increases indefinitely, child mortality approaches its asymptotic value of about 82 deaths per thousand. As explained in footnote 20, the positive value of the coefficient of $(1/PGNP_i)$ implies that the rate of change of CM with respect to PGNP is negative.

FUNCTIONAL FORMS OF REGRESSION MODELS

Model	Equation	Slope $\left(= \frac{dY}{dX} \right)$	Elasticity $\left(= \frac{dY}{dX} \frac{X}{Y} \right)$
Linear	$Y = \beta_1 + \beta_2 X$	β_2	$\beta_2 \left(\frac{X}{Y} \right)^*$
Log-linear	$\ln Y = \beta_1 + \beta_2 \ln X$	$\beta_2 \left(\frac{Y}{X} \right)$	β_2
Log-lin	$\ln Y = \beta_1 + \beta_2 X$	$\beta_2 (Y)$	$\beta_2 (X)^*$
Lin-log	$Y = \beta_1 + \beta_2 \ln X$	$\beta_2 \left(\frac{1}{X} \right)$	$\beta_2 \left(\frac{1}{Y} \right)^*$
Reciprocal	$Y = \beta_1 + \beta_2 \left(\frac{1}{X} \right)$	$-\beta_2 \left(\frac{1}{X^2} \right)$	$-\beta_2 \left(\frac{1}{XY} \right)^*$
Log reciprocal	$\ln Y = \beta_1 - \beta_2 \left(\frac{1}{X} \right)$	$\beta_2 \left(\frac{Y}{X^2} \right)$	$\beta_2 \left(\frac{1}{X} \right)^*$

Note: * indicates that the elasticity is variable, depending on the value taken by X or Y or both. When no X and Y values are specified, in practice, very often these elasticities are measured at the mean values of these variables, namely, $\bar{X}$ and $\bar{Y}$.

Choice of Functional Form: Some Practical Considerations

Some guidelines are:

- Underlying theory/hypothesis
- Coefficients should satisfy prior expectations
- Model that fits well to data is better
- Don't over-emphasize R square as a criterion to judge which model is better
- Don't compare R squares of the two models if dependent variable is not same
- Three types of significance, namely economic, statistical and econometric significance may be taken care of.